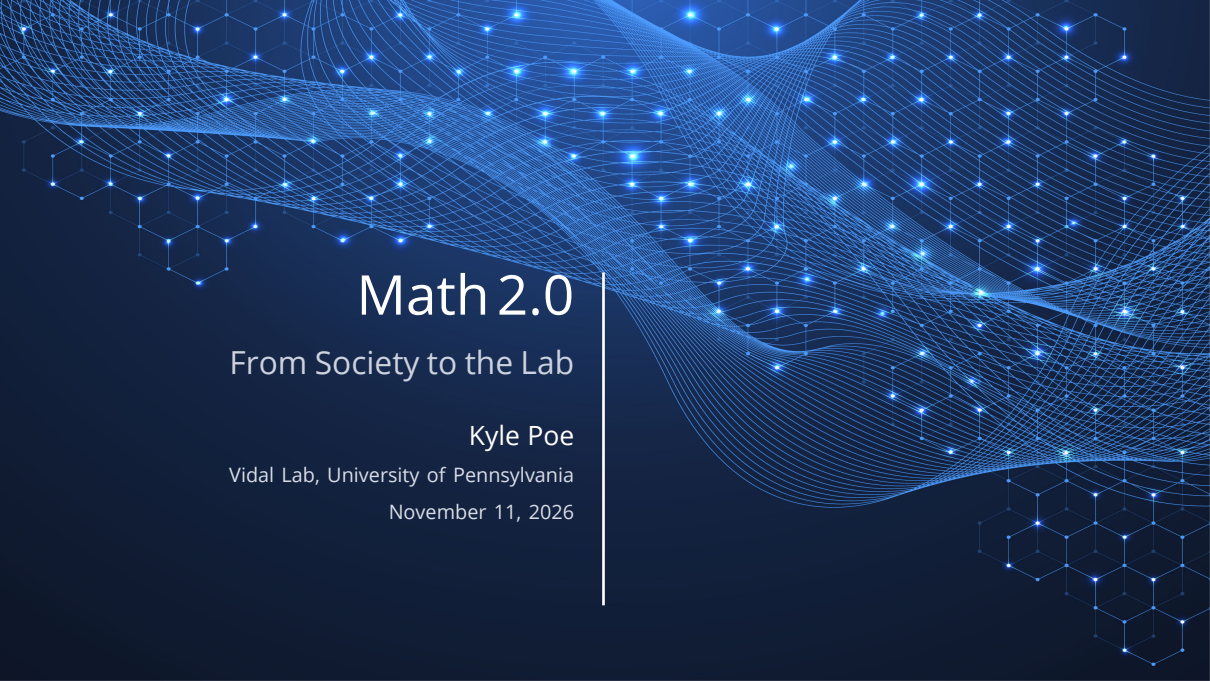


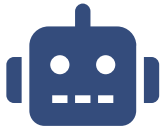
Math 2.0

From Society to the Lab

Kyle Poe

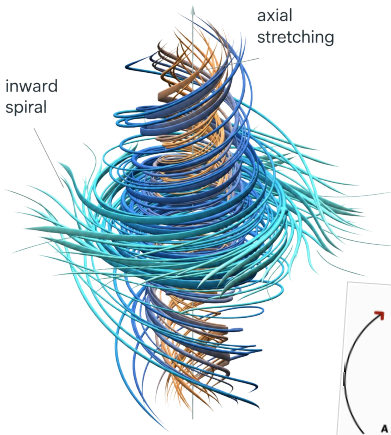
Vidal Lab, University of Pennsylvania

November 11, 2026



This presentation was prepared with the assistance of Claude Opus 5.5

September 8, 2026: Math changes forever



OpenAI @OpenAI

We're sharing a solution to the Navier-Stokes Millennium Prize Problem, one of the deepest problems at the frontier of mathematics.

nature

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nature › news › article

NEWS | 08 September 2026

OpenAI claims huge maths
gh on a famed
n Problem'

cked the physics of fluids
in mathematics.

OpenAI Claims to Have Solved a $1M Math Problem: A Credit Dispute Has Already Begun

Ten thousand AI agents resolved one of math's oldest open problems in 88 hours. A credit dispute over authorship erupted almost immediately.

BY TECHNOLOGY DESK · SEPTEMBER 9, 2026 · 3 MINS READ

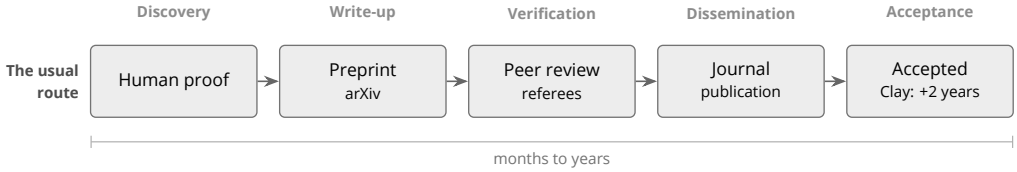


OpenAI Cracks 200-Year-Old 'Millennium' Math Problem—But Credit Dispute Looms

An NYU math professor claims the AI behemoth rushed to solve the problem after discovering his progress.

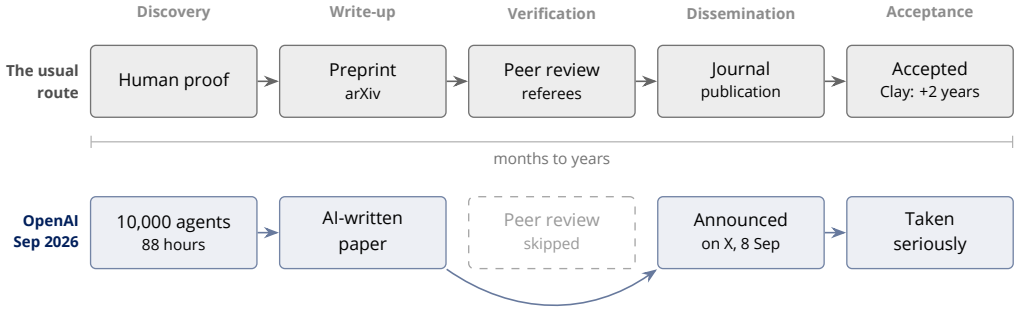
Sources: OpenAI on X, 8 Sep 2026 • Nature news, 8 Sep 2026 • Technology Desk, 9 Sep 2026 • Forbes • Visualization: OpenAI • Meme: @MozeTech on X

Why believe it?



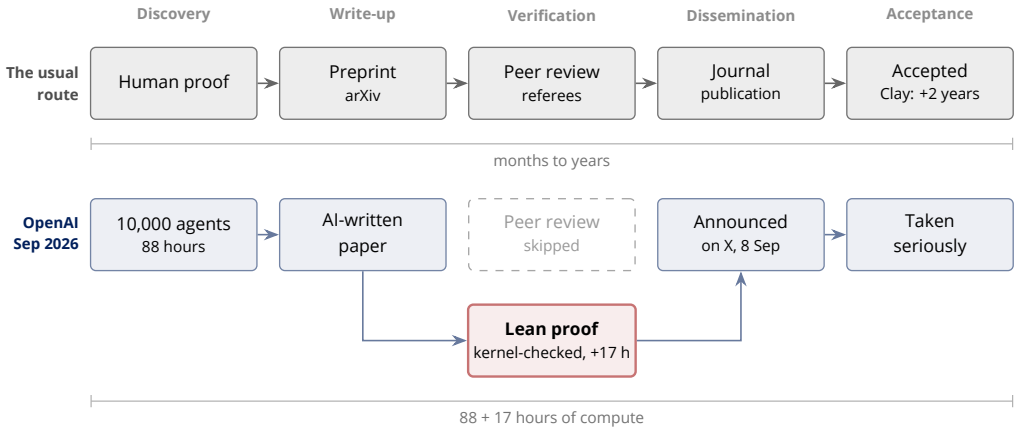
Sources: OpenAI, 8 Sep 2026 • Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, Oct 2026 • Clay Mathematics Institute, Millennium Prize rules

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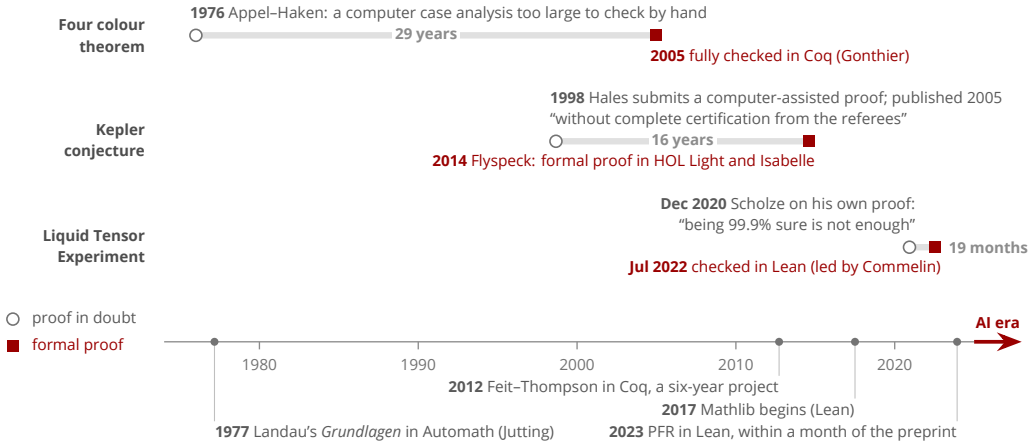
Why believe it?



Lean verification, not merely OpenAI's reputation, made the community take this seriously.

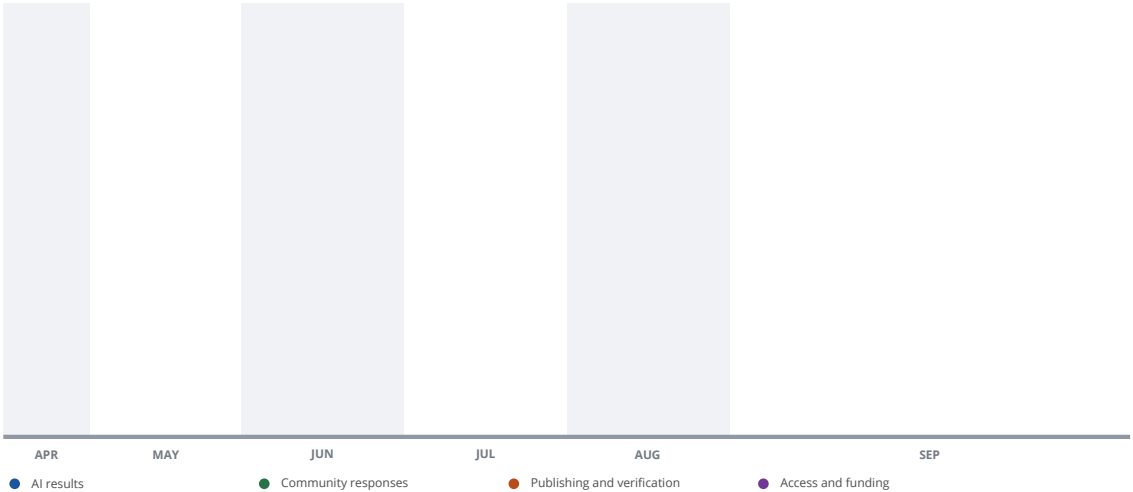
Sources: OpenAI, 8 Sep 2026 • Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, Oct 2026 • Clay Mathematics Institute, Millennium Prize rules

Timeline: Formalization in the Pre-AI Era



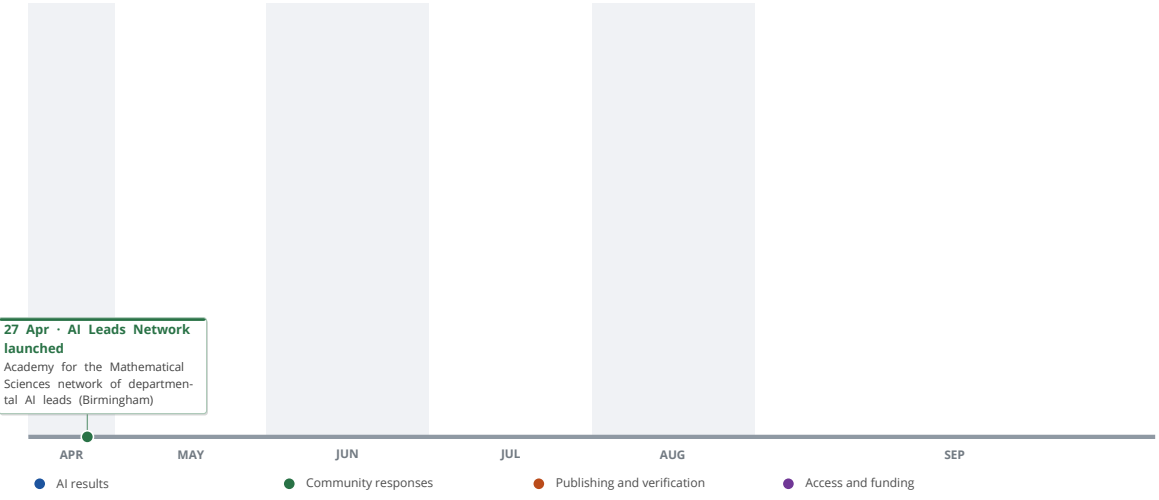
Sources: Jutting, thesis (1977) • Gonthier, MSR report (2005) • Hales, Annals 162 (2005); Hales et al., arXiv:1501.02155; Flyspeck repo • Gonthier et al., ITP 2013 • Scholze, Xena blog (Dec 2020); Lean blog (Jul 2022) • mathlib Community, CPP 2020 • Gowers et al., arXiv:2311.05762; Tao, blog (Dec 2023)

Timeline: AI Transforms the Mathematical Sciences



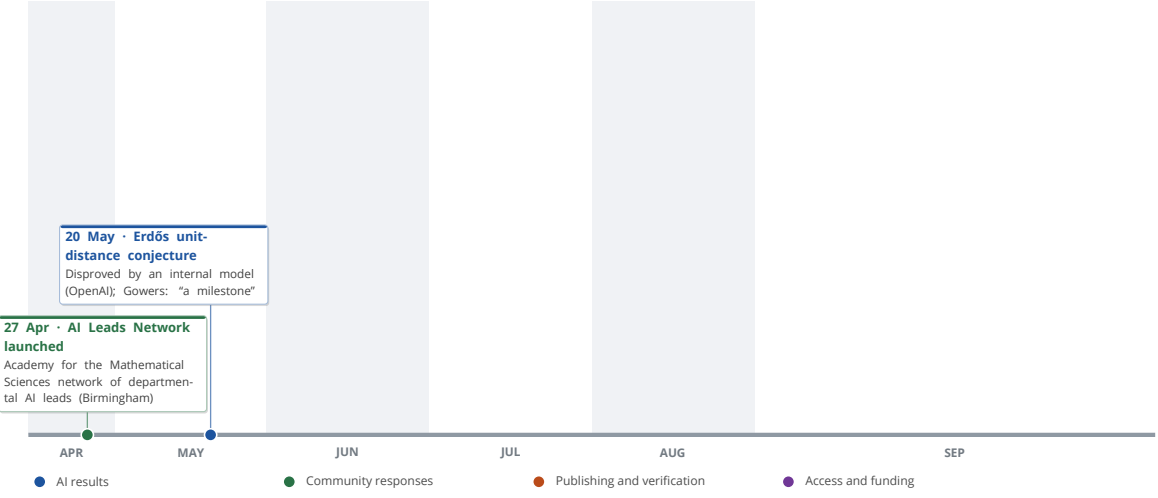
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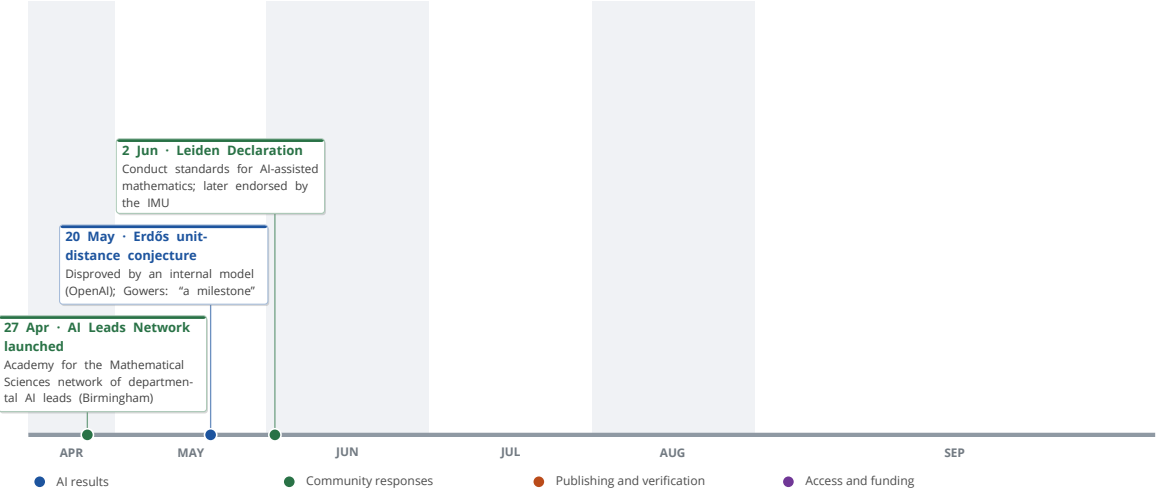
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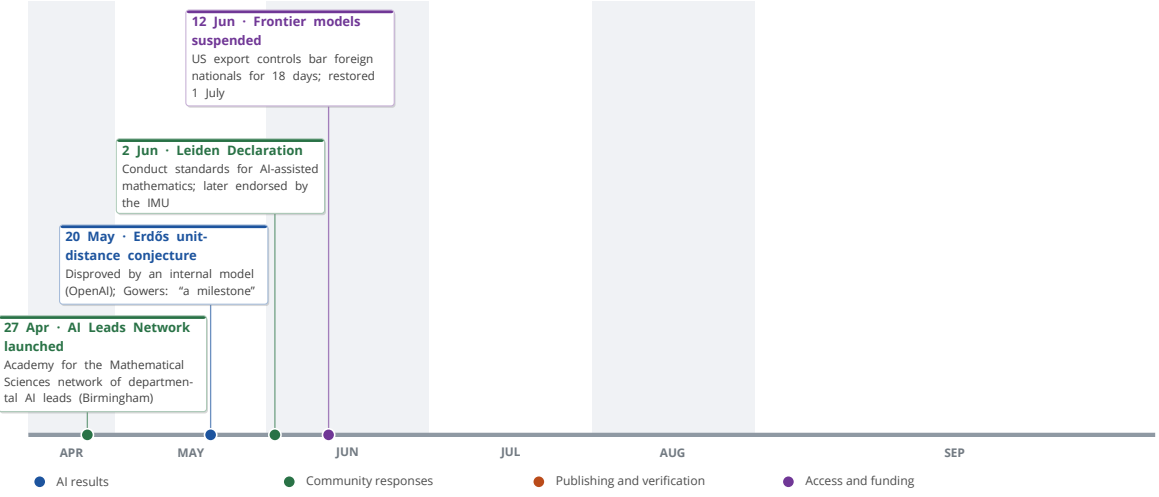
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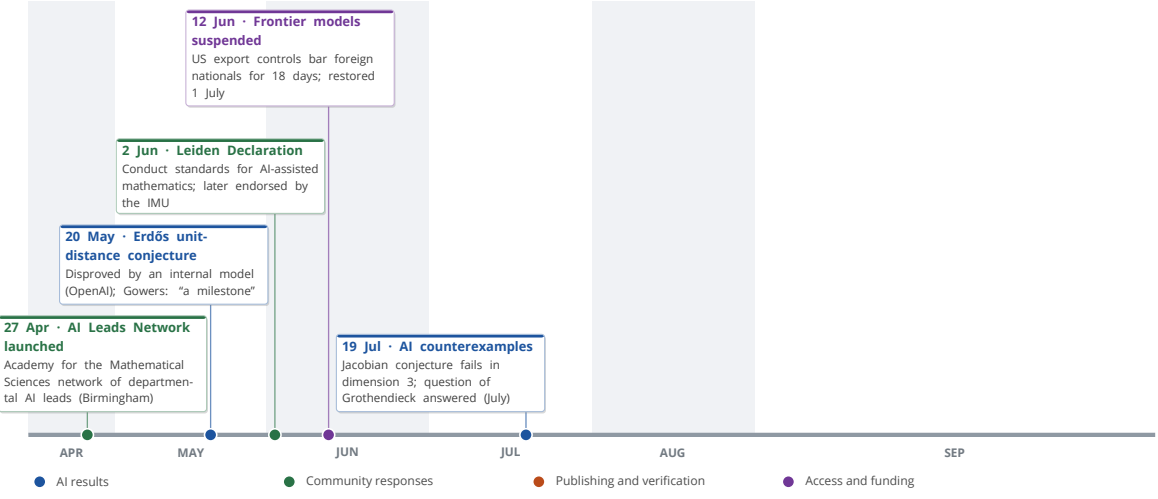
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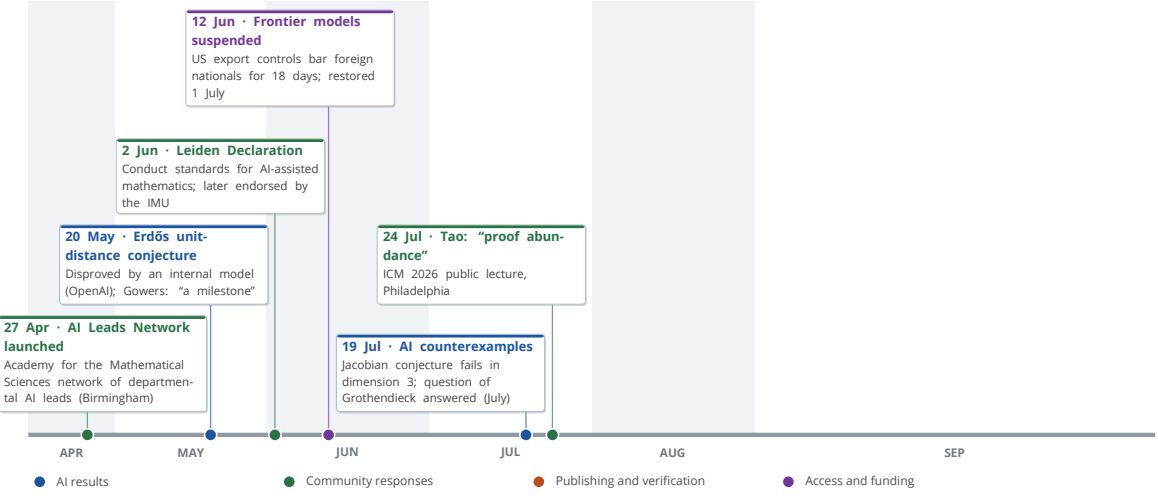
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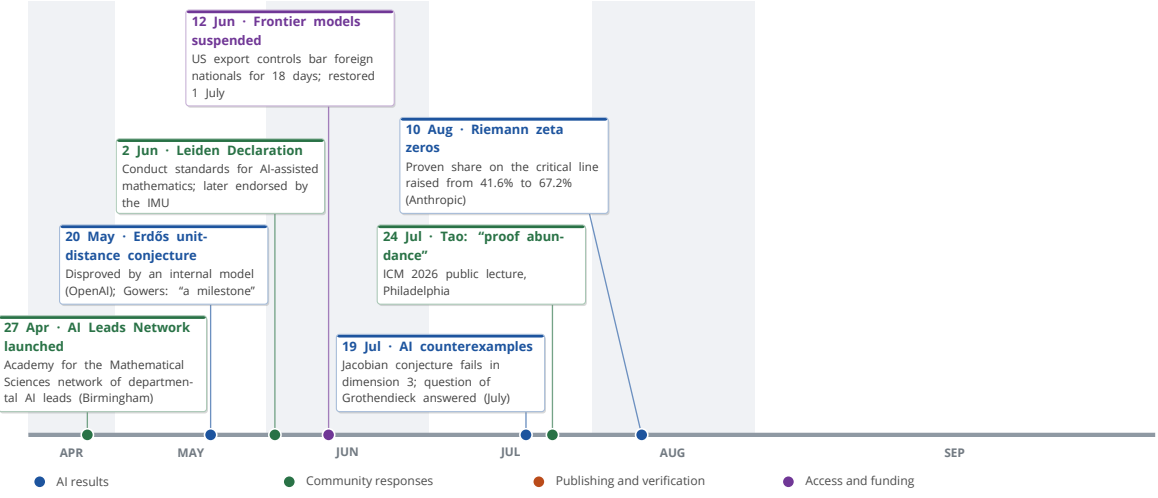
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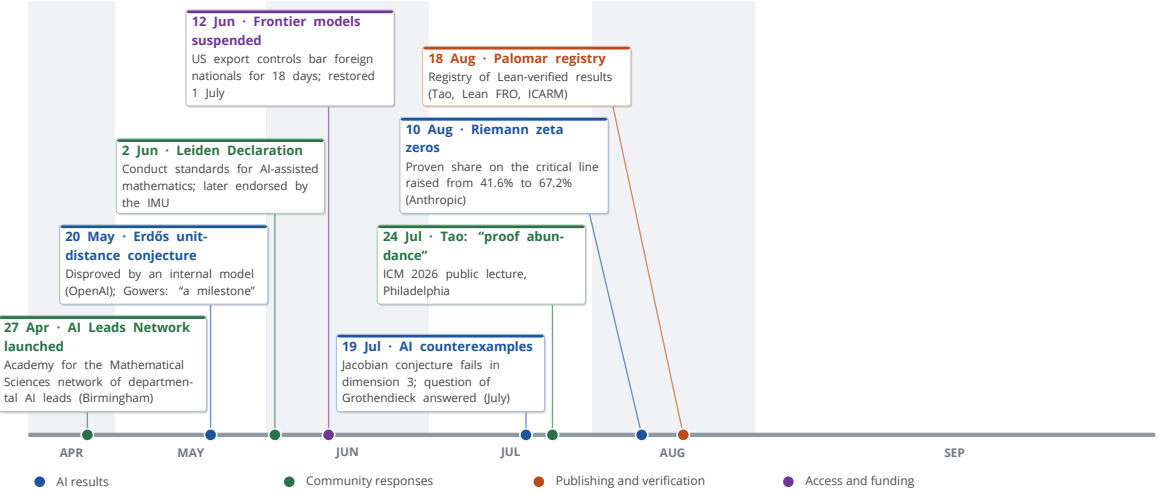
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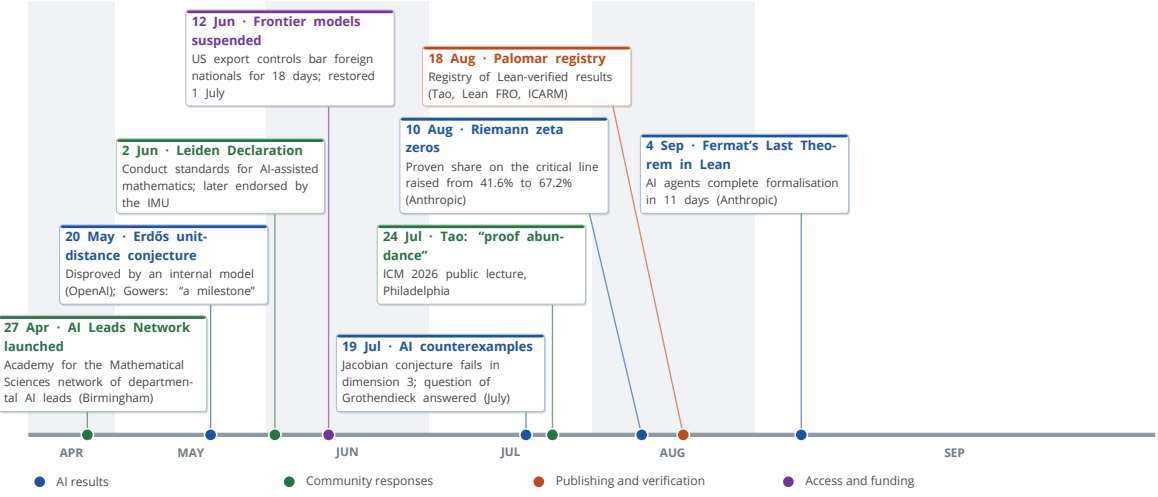
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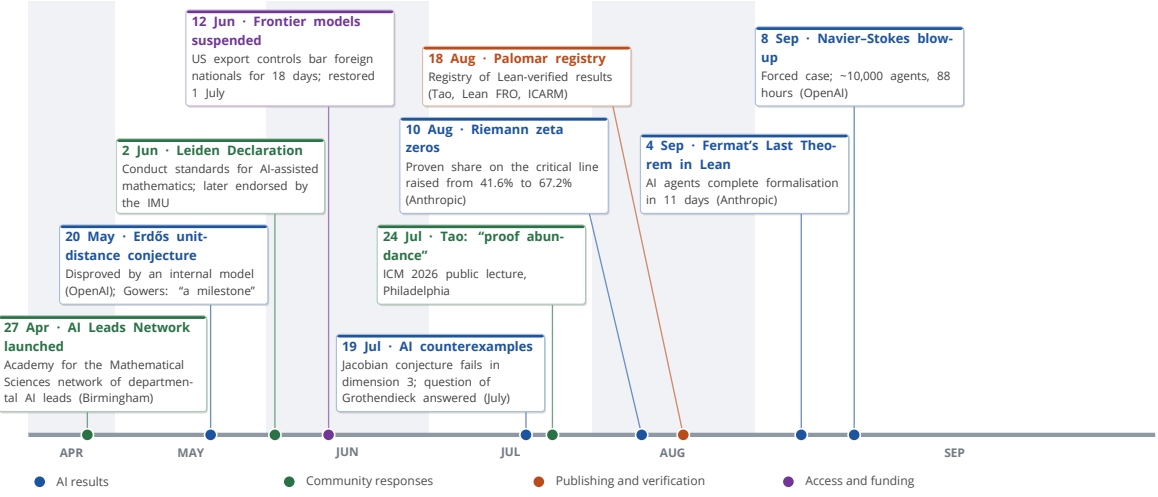
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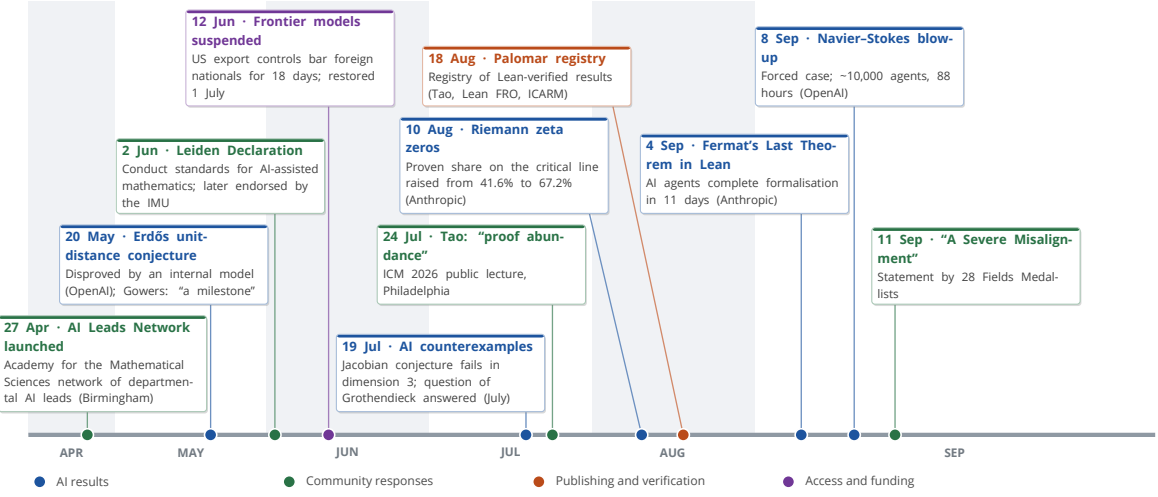
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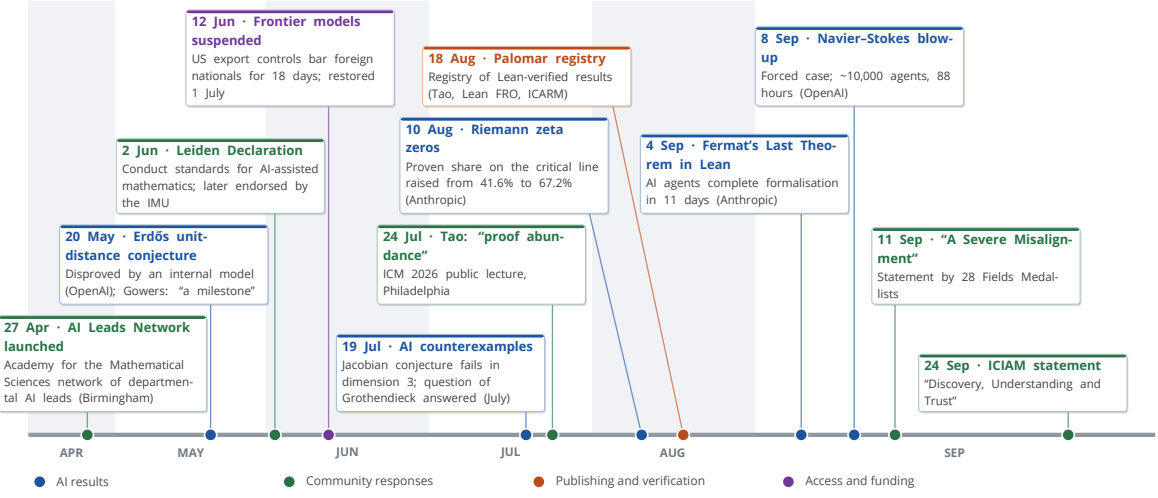
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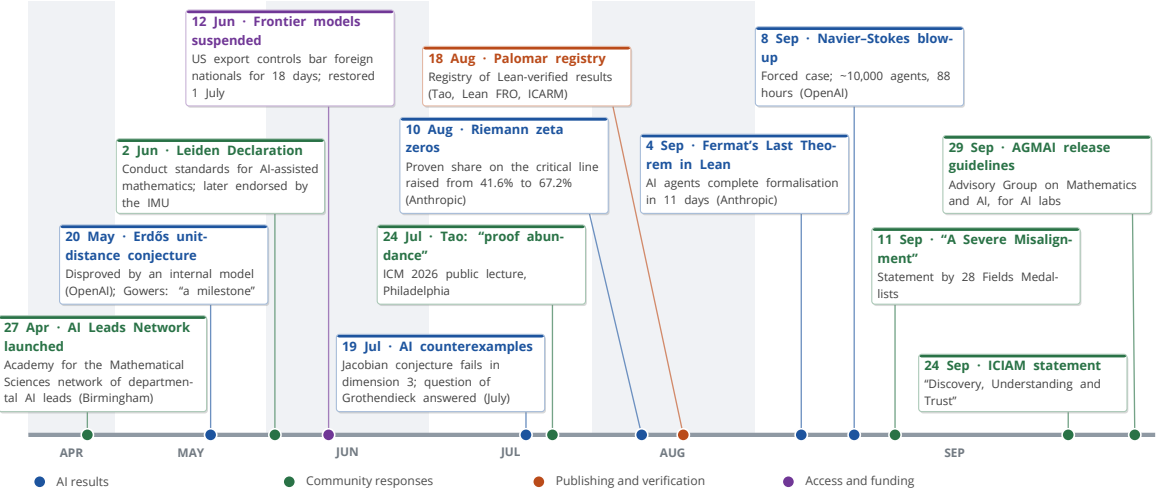
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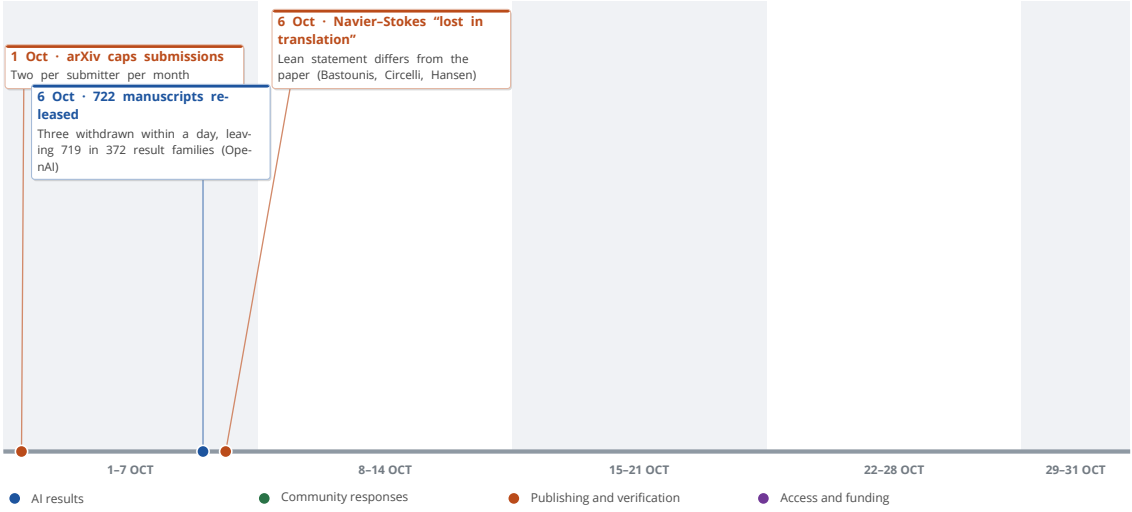
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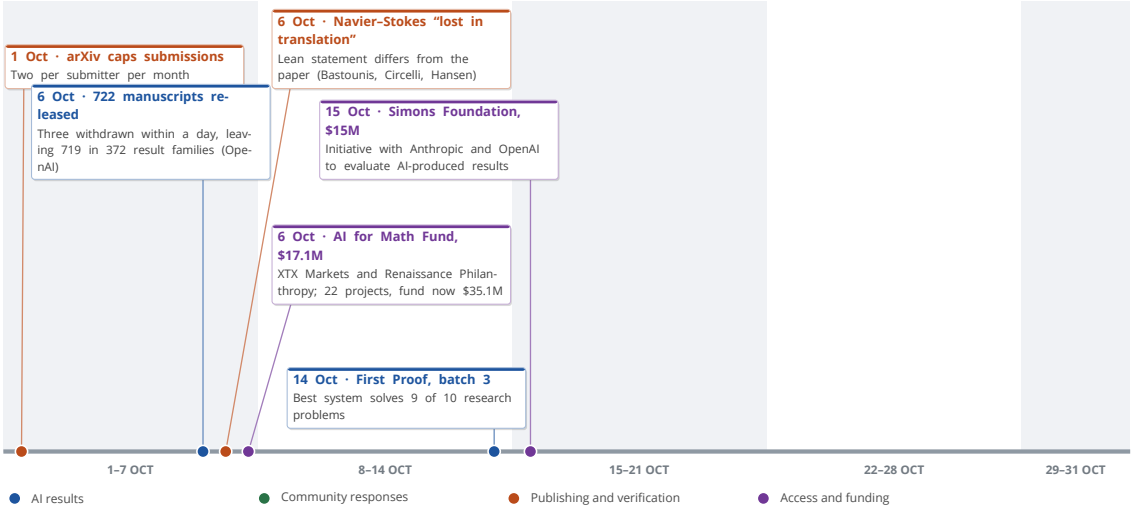
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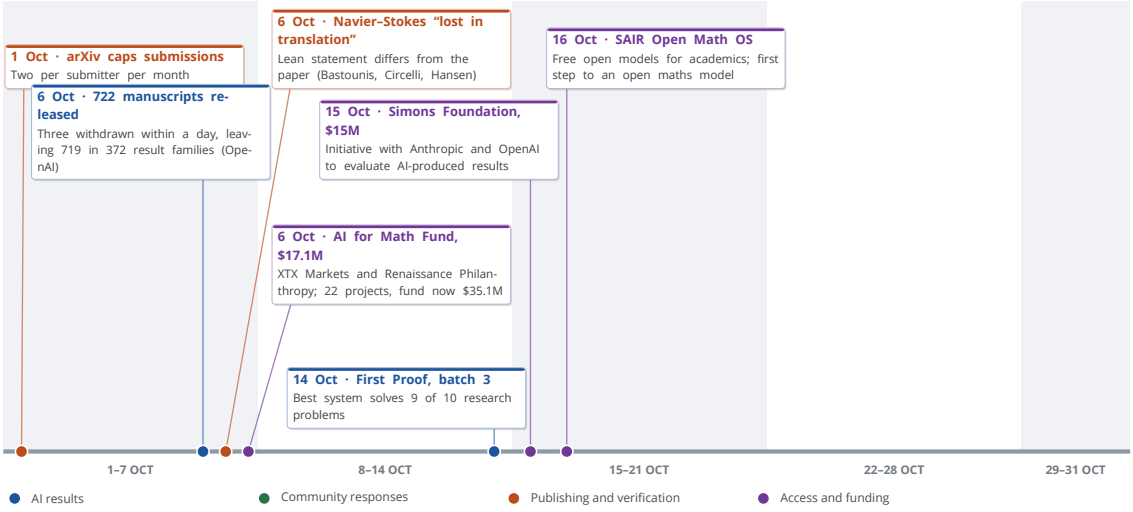
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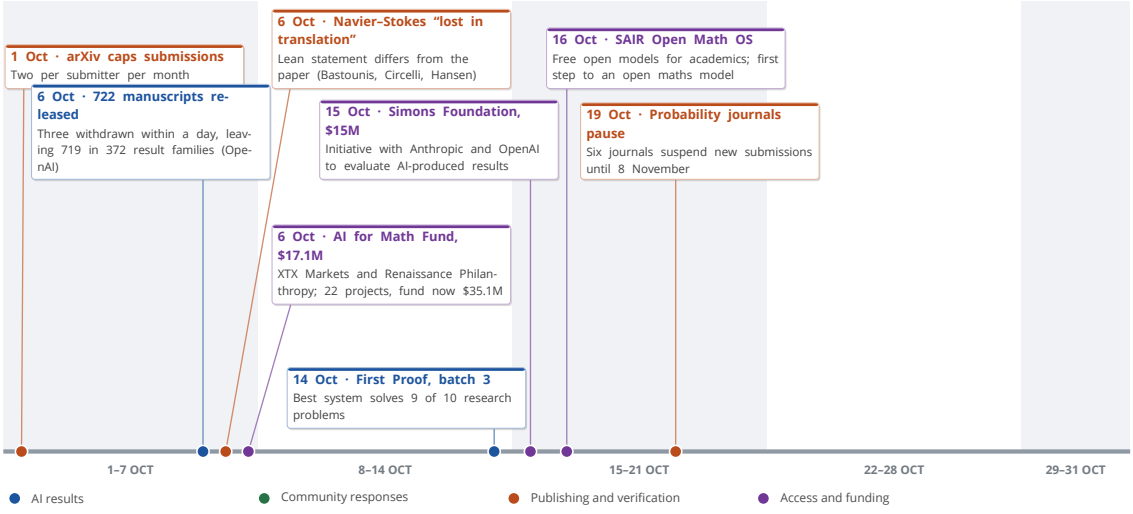
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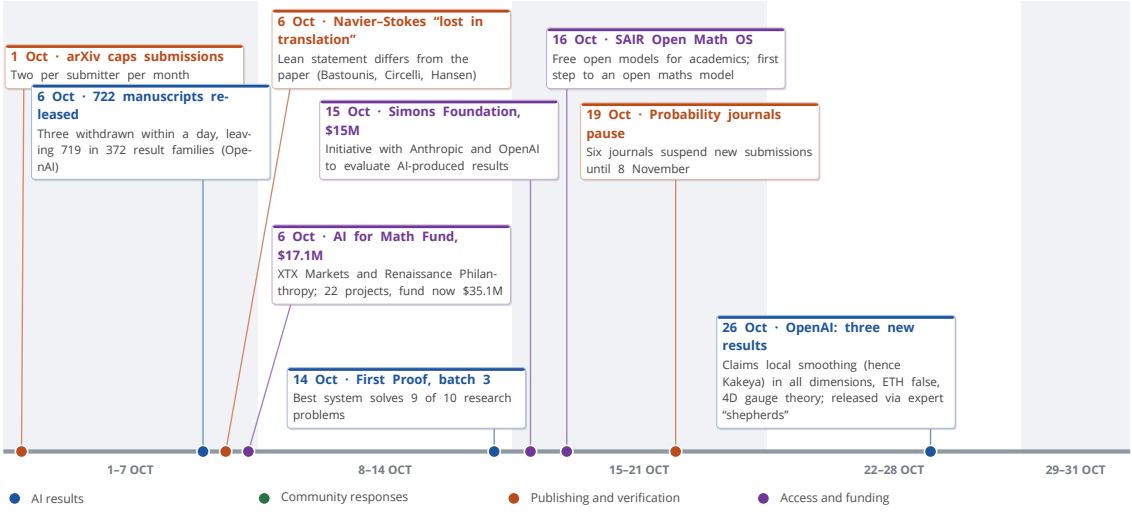
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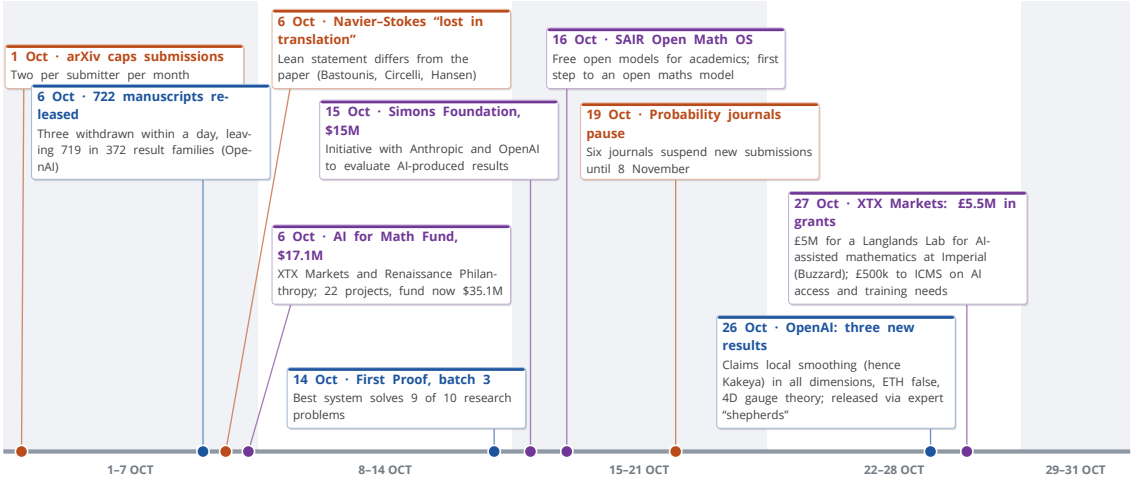
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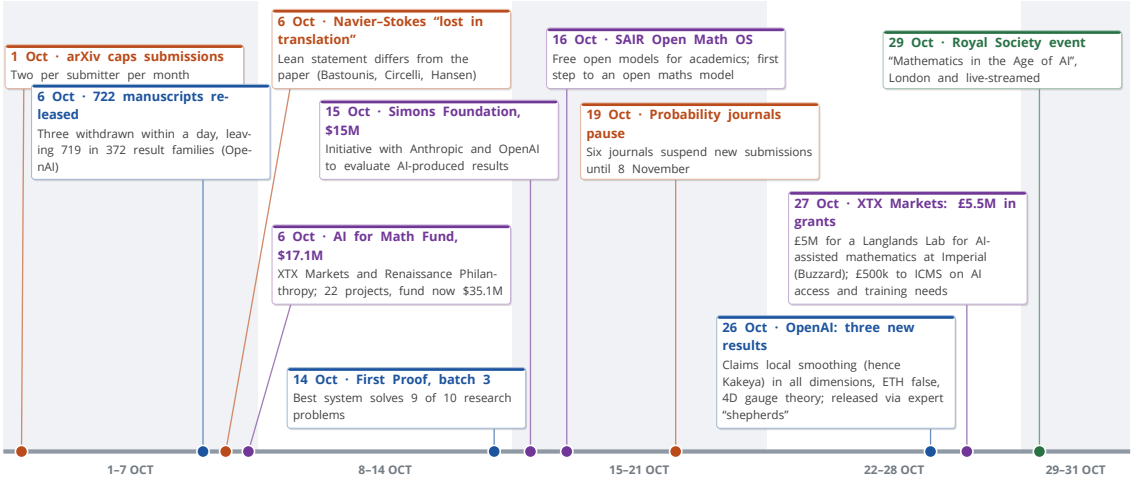
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Math 1.0 and Math 2.0

	Light AI use	Heavy AI use
Weak human understanding	**Pre-math**	**Misaligned math**
Strong human understanding	**Math 1.0**	**Math 2.0**



> [*Rather than blindly using AI to generate misaligned mathematics,*] *the more sustainable* "Math 2.0" *use of AI would be to strengthen all the ways in which human understanding interacts with mathematics.*
>
> — Terence Tao, *Machine Assisted Proof*,
> AIM Public Lecture, Caltech, 9 Oct 2026

Quote and diagram: T. Tao, *Machine Assisted Proof*, AIM Public Lecture Series, Caltech, 9 Oct 2026 (caltech.edu/campus-life-events/calendar/aim-public-lecture-series-terence-tao-1)

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Amidst the plethora of ethical concerns surrounding proliferation and normalization of the technology, AI is ushering in a new dominant paradigm for mathematical, and more broadly theoretical, research. Thoughtfully deployed, AI-assisted formalization tools will be the glue between human and machine understanding which ensures the integrity and alignment of future scientific enterprise.

Quote and diagram: T. Tao, *Machine Assisted Proof*, AIM Public Lecture Series, Caltech, 9 Oct 2026 (caltech.edu/campus-life-events/calendar/aim-public-lecture-series-terence-tao-1)

The rest of this talk

PART I
In Society

- Public reactions to Math 2.0
- Open Societal Problems
- Formalization
- The Limits of Formalization

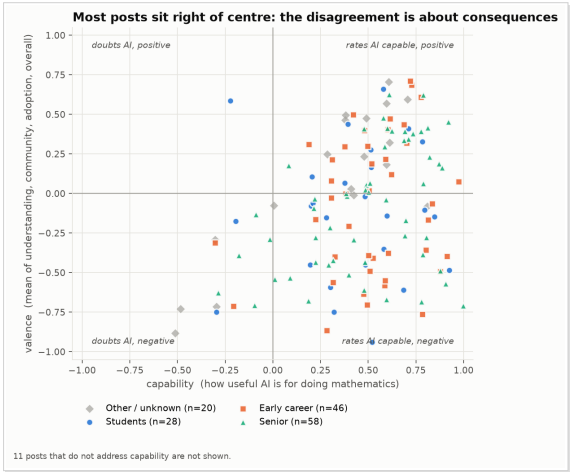
PART II
In The Lab

- Tools and workflow
- The Lean Blueprint
- Statement Fidelity
- Proof Fidelity

Part I

Society

Proofs and Prompts: A Sentiment Analysis





Proofs and Prompts

"Notes from the common room"

A blog by mathematicians, for mathematicians, on how AI is changing their practice. 171 posts from 151 contributors, Aug–Nov 2026. No party line; it invites PhD students and skeptics.

152 essays, each scored by an LLM

89%

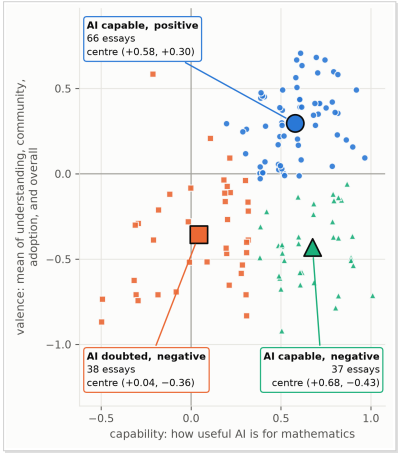
rate AI as capable of doing mathematics

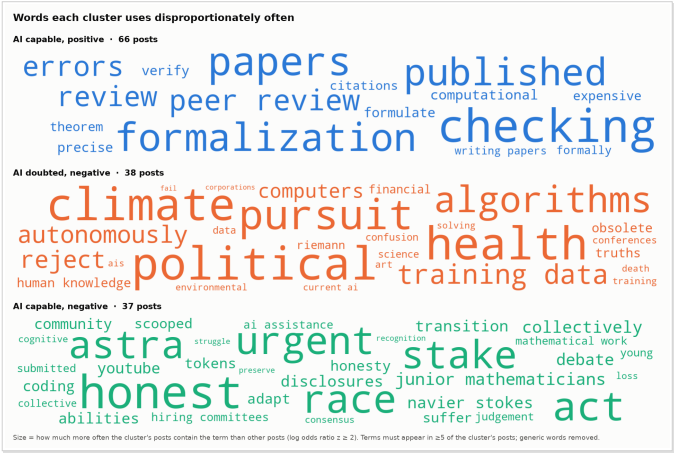
49%

view AI as a negative force overall

proofsandprompts.com, About page (logo by Radhika Gupta); posts crawled 7 Aug–9 Nov 2026 • Proofs & Prompts corpus, scored with Claude; analysis set of 152 posts, 141 addressing capability. Self-selected essays: an illustration, not a survey

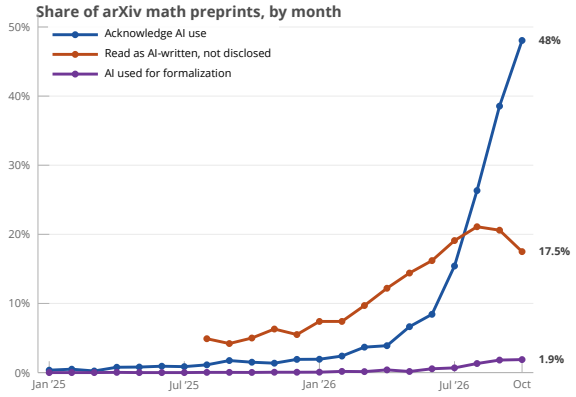
Proofs and Prompts: What Each Cluster Writes About





Proofs & Prompts corpus; scores from Claude (rubric.py); clustering and word statistics from cluster.py. Self-selected essays: an illustration, not a survey

The Poison Fountain



1. **Use is exploding.** Disclosed AI use went from 0.4% to 48% of math preprints in 21 months, doubling about every three months. Math submissions are up 75% on a year earlier (Sep 2026).

2. **Much of it is undisclosed.** About 1 in 6 math preprints read as AI-written without saying so (Martin, Oct 2026). At ICLR 2026, 9% of submissions were mostly AI-written.

3. **Few certify.** OpenAI shipped Lean proofs with 42% of its 719 manuscripts. Across arXiv math, 1.9% acknowledge AI for formalization; in stat.ML, 0.4% even mention Lean.

4. **Review is overwhelmed.** 21% of ICLR 2026 reviews were fully AI-generated. arXiv capped submissions (1 Oct); six probability journals paused (19 Oct – 8 Nov).

Epoch AI (CC BY 4.0, 9 Nov 2026) • K. Martin, *AI use in arXiv mathematics papers*, kimbalmartin.github.io/ai-use (3 Nov 2026, 17.5% of Oct 2026 math preprints) • our arXiv count • Pangram, ICLR 2026 (self-reported) • AcadMathSci, Oct 2026

Credit and careers

Problem	Case	Response	Status
Credit	AI reuses known ideas without attribution: 8 of Aletheia's 13 meaningful Erdős results were already in the literature. Kolda: "Human–AI scrapple"	Leiden Declaration (Jun 2026, endorsed by the IMU): human responsibility, attribution	• partial
Priority	Navier–Stokes priority dispute (Córdoba & Martínez-Zoroa; Buckmaster & Alpöge). A result named in a talk title was pre-empted by a hasty AI-generated paper (anecdotal)	AGMAI release guidelines (29 Sep): review before announcement, release prompts. Labs withheld prompts	• open
Paper mills	A company could solve open questions with AI and sell the authorship of correct papers	None yet	• open
Careers	**"I no longer trust the profession to offer a fair and intelligible path forward"** (anonymous PDE postdoc)	Judge a few best outputs; viva-style questioning for hiring (AcadMathSci)	• partial

Reisinger et al., AcadMathSci, Oct 2026 (§3.2, §3.4, §4.4, A.3, C.1) • Kolda, Silicon Reckoner, Feb 2026 • Proofs & Prompts posts 1043 (Dolores-Cuenca) and 1893 (anonymous)

Understanding

Problem	Case	Response	Status
Deskilling	Students given unrestricted GPT-4 scored 17% lower once access was removed; a tutor with guardrails largely avoided the harm (Bastani et al.)	Vivas and in-person exams; AI-permitted work graded on formulation and validation	• partial
Narrowing	AI adoption shrinks topic range by 4.6% and engagement by 22% (Hao et al., 41M papers). Agendas drift toward "problems easily amenable to current AI"	Reward understanding and long-term contribution over volume (AcadMathSci)	• open
Opaque output	Hairer: AI proofs detail the easy steps and skim the hard ones, and are hard to trust. Moshkovitz, on one of OpenAI's papers: impossible to read without AI help	Leiden: preserve human understanding. Proposals to publish the certificate apart from the explanation	• open

Bastani et al., PNAS • Hao et al., arXiv 2412.07727 • Reisinger et al., AcadMathSci, Oct 2026 (§3.2, §3.3) • Hairer, NYT interview, Feb 2026 (secondary) • Leiden Declaration, Jun 2026

Ongoing, without clear solutions

Integrity		Credit and careers		Understanding	
Volume of submissions	• partial	Credit	• partial	Deskilling	• partial
Undisclosed AI	• partial	Priority	• open	Narrowing	• open
Overwhelmed review	• open	Paper mills	• open	Opaque output	• open
		Careers	• partial		

None of these ten problems has a solution in place.

The one benefit that comes up across the board: formalization

36% of the essays in **AI capable, positive** discuss it

27% of the essays in **AI capable, negative** discuss it

29% of the essays in **AI doubted, negative** discuss it

47 of 152 essays mention formalization, Lean or proof assistants. In the most positive cluster, *checking*, *formalization*, *errors* and *verify* are among the most distinctive words.

The three slides above • Proofs & Prompts corpus: keyword count over the 152-post analysis set; word statistics from cluster.py

Formal verification has advocates in all three groups

AI capable, positive

"Proofs should be Lean-formalized where possible, unless it leads to unacceptable delays."

AGMAI, release guidelines

"Mathematicians will use Lean formalization and verification of proofs, but will also complement it with human-readable exposition."

Etingof & Gerovitch, MIT

AI capable, negative

"I expect verifying correctness will soon become a non-issue through a combination of better AI and Lean verification."

Harlev, Cornell

"… an AI that formalizes the proof can further filter the papers."

Dolores-Cuenca, Yonsei, on paper mills

AI doubted, negative

"… we might be able to solve the problem (that of the convincing incorrect argument) because we have the idea of a proof, … and we can insist on formalization."

Weinberger, Chicago

"… many mathematicians would begrudgingly accept its correctness if some AI agent produces suitable Lean code tomorrow."

Nesterov, Stanford, on the finite simple groups

Where they push back

"In mathematics, we do judge submissions for non-scientific factors that concern plagiarism, ownership, and originality."

Kretschmer, UT Austin, against judging papers by their Lean certificates alone

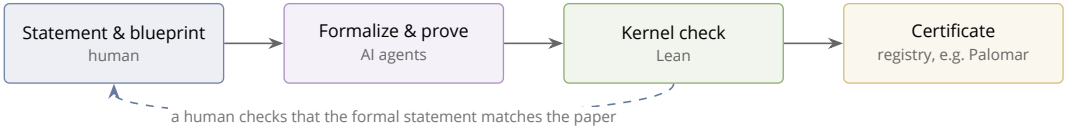
Their initial goal was "to formalize mathematics in a way that makes it globally understandable."

Stiegler, Paris-Saclay, on friends who work on Lean and are unhappy with where it is heading

Proofs & Prompts posts 3065, 2114, 757, 1043, 146, 830, 2379, 2020; clusters as on the sentiment slides

AI-Assisted Formalization in Practice

AI translates mathematics into a proof assistant such as Lean and fills in the proofs. A small trusted kernel checks every step. Humans write, or review, what is being proved.



Already being built

Palomar

A registry of Lean-verified mathematics (Lean FRO and ICARM, Aug 2026). Two independent kernels replay every proof.

Comparator

Checks that a proof proves exactly the stated theorem, with no extra axioms. Part of Lean since v4.35; used for FLT.

New venues

A journal that requires Lean-verified proofs (Guo, proposed Oct 2026). A STOC/FOCS/SODA track where a Lean certificate stands in for the proof (proposed).

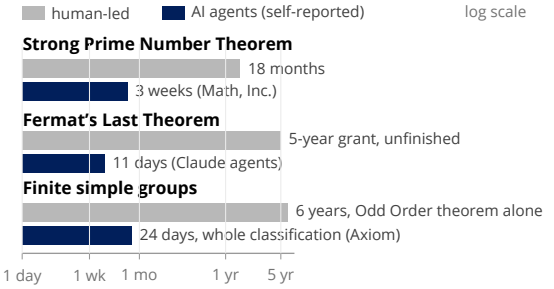
Lab releases

OpenAI shipped Lean proofs with 42% of its 719 manuscripts. FLT and the finite simple groups shipped challenge files.

Tao, *Palomar*, blog, 18 Aug 2026 • Lean FRO, Comparator • Guo, guest post on Tao's blog, 17 Oct 2026 • Kothari, Meka & Raghavendra, Shtetl-Optimized, 19 Sep 2026 • research notes 01–03

Cheap, and it catches errors

Time and cost to formalize



$200 a page

Grinberg's 500-page textbook: all 340 targets in about a week, about $100k (Meta)

£91–226k

Fermat's Last Theorem: about 6B output tokens (AcadMathSci estimate)

5–10×

cheaper each year for a fixed level of capability

Errors found by formalizing

OpenAI's 719 manuscripts

3 withdrawn within a day after a sign error; **all three were outside the Lean coverage**. An independent audit of the 245 formalized families found no theorem shown false.

Finite simple groups

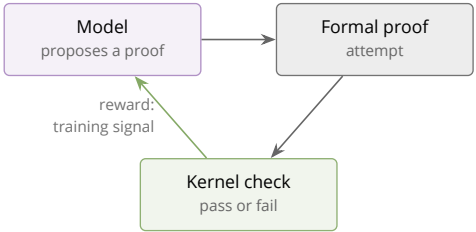
34 deviations from the published record, including a false equality in a published quasithin lemma (a counterexample in $SL_2(5)$). Axiom, self-reported; the build has not been reproduced.

The Stacks Project

Dozens of statements that fail in the stated generality: missing hypotheses, edge cases in positive characteristic. Axiom, self-reported.

Math, Inc., Strong PNT (2025) • Anthropic, FLT (Sep 2026) • Meta, arXiv 2604.03071 • AcadMathSci, Oct 2026 (B.1, B.3) • Axiom, CFSG (Oct 2026) and Stacks (Nov 2026) • Gonthier et al., ITP 2013 • bolt/sm/openai-math-slop, Issue #2 • research note 01

No verifier, no flywheel



AlphaProof autoformalized about 1M problems into about 80M Lean statements to train on this loop.

- Where a result can be checked automatically, every attempt is a training signal. AcadMathSci: Lean is becoming the way AI checks its own work.
- Where it can't (modelling choices, significance, exposition), a person has to judge every output.
- So capability "will stay uneven across the mathematical sciences".

Where formalization lies, AI-driven gains follow.

Reisinger et al., AcadMathSci, Oct 2026 (§1.2; A.5, the report's assessment)

But when can it be trusted?

For an AI-generated Lean certificate to be useful, it must satisfy three key criteria: **validity**, **statement fidelity**, and **proof fidelity**.

1. Validity

Does Lean accept the proof?

2. Statement fidelity

Does the formal statement say what the paper claims?

3. Proof fidelity

Does anyone understand the proof, and does it match the written one?

Research note 01, §0

Formal certificates have failed us before

1. Validity

The Collatz "disproof", Jul 2026

- A Lean repository with no `sorry` "disproved" the Collatz conjecture.
- It passed Lean's kernel and an independent checker. It was built on purpose, from a kernel bug that allowed a proof of *False*.
- Fixed about an hour after it was reduced to a proof of *False*. The advisory covers every Lean 4 version up to v4.31.

2. Statement fidelity

Proved, but not what was meant

- An OEIS conjecture "disproved" by AI for $0.17 of API spend. The formal statement was wrong.
- Erdős problem #124 was restated after AI attempts showed it was unclear which density was meant.
- Lean's conventions can change a meaning: the infimum of an empty set of naturals is 0.

3. Proof fidelity

Correct, but not the written argument

- Navier–Stokes: the paper's estimate needs $m+4$ derivatives; the Lean lemma assumes $m+5$. A pressure bound depends on an extra quantity and takes another route.
- Fermat's Last Theorem: about 13M lines, "likely far longer than necessary".

"Lean acceptance (compilation) of translation does not imply semantic preservation."

— Bastounis, Circelli & Hansen, on the Navier–Stokes formalization

Lean 4 security advisory, oss-sec, 2 Aug 2026 • google-deepmind/formal-conjectures, issue #5226 • Xena blog • Bastounis, Circelli & Hansen, arXiv 2610.08144 • AcadMathSci, Oct 2026 (A.2) • research notes 01–02

What certificates settle, and what they don't

Problem	What formalization does
Review overwhelmed	The kernel checks every step; referees read statements, not proofs. Only where the libraries reach
Undisclosed AI	The check reads the proof, not the author
Statement fidelity	A person still has to confirm the formal statement says what was meant
Proof fidelity	A correct machine proof can still be unreadable, or follow a different argument
Credit and careers	Registries record what produced a result. Who had the idea is not a formal property
Deskilling and narrowing	Nothing directly
Funding, access, energy	Nothing directly. Only 11% of researchers have institutional funding for their main AI tool

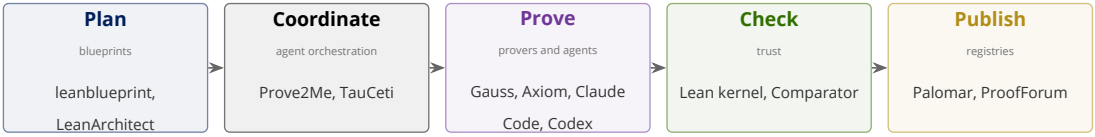
What does a principled deployment of Math 2.0 look like?

Sections above • Springer Nature survey of 10,480 researchers, via AcadMathSci, Oct 2026

Part II

The Lab

The stack



Libraries Mathlib • TauCeti • PhysLib • CSLib • EconCSLib everything above builds on these

In and around learning theory

already in Lean 4: not as far away as it may seem

SLT (Zhang, Lee & Liu, ICML 2026)

- Dudley's entropy integral, chaining
- Gaussian log-Sobolev, Efron–Stein
- Hanson–Wright, matrix Bernstein
- Davis–Kahan perturbation bounds
- Minimax rates for ℓ_1 regression

StatsMLlib (65k lines, 666 theorems)

- Concentration of measure
- Random matrix theory
- Rademacher complexity, metric entropy
- Finite-sample learning guarantees

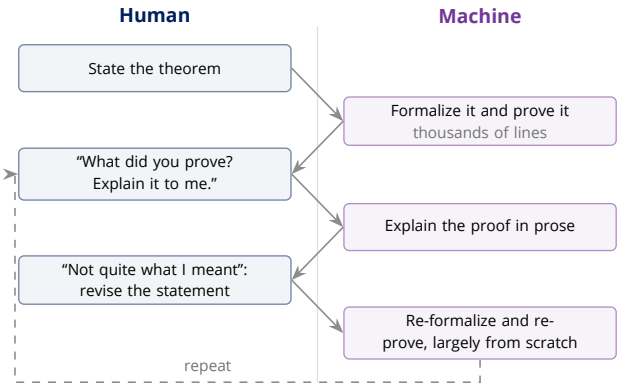
lean-rademacher: Dudley's bound

CvxLean (Bentkamp et al., TACAS 2023)

- Convex problems written in Lean
- Verified reductions between problems
- Proof-producing solver canonization

Research note 02 • Zhang, Lee & Liu, AI4SLT, arXiv 2602.02285 (github.com/YuanheZ/lean-stat-learning-theory) • statsmllib.github.io • Bentkamp, Fernández Mir & Avigad, TACAS 2023 (github.com/verified-optimization/CvxLean)

The naive workflow



Why this goes wrong

- The two sides only ever agree on the whole theorem, so every change to the statement restarts the proof.
- The human checks a machine’s summary of a machine’s proof, with no shared structure to hold it against.
- Statement errors surface last, after the compute is spent.
- Nothing persists but the chat: no reusable lemmas, no record of what was checked.

In Anthropic’s formalization of Fermat’s Last Theorem, attempts made without a dependency graph contributed only about 7% of the final lines (Prove2Me).

The human and the machine never agree on anything smaller than the whole theorem.

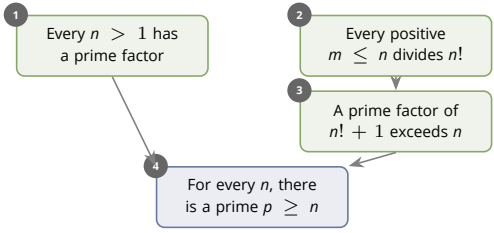
Prove2Me, arXiv 2608.28433, and Anthropic, FLT (Sep 2026), via research note 02

Blueprints

- The naive loop negotiates one statement and one proof. A blueprint replaces them with a **dependency graph**: every definition and lemma is a node, stated in words and in Lean.
- Agree on the statements first, with proofs left open. Then fill in the proofs node by node.
- A revised statement touches one node and what depends on it, not the whole proof.
- Progress is visible: each node is stated, ready, or proved.
- **leanblueprint** turns the paper into the graph; **LeanArchitect** builds the graph from Lean code; **Verso Blueprint** does both inside Lean. Used for Liquid Tensor, PFR, FLT, sphere packing and Carleson.

The statement dependency graph becomes the base unit.

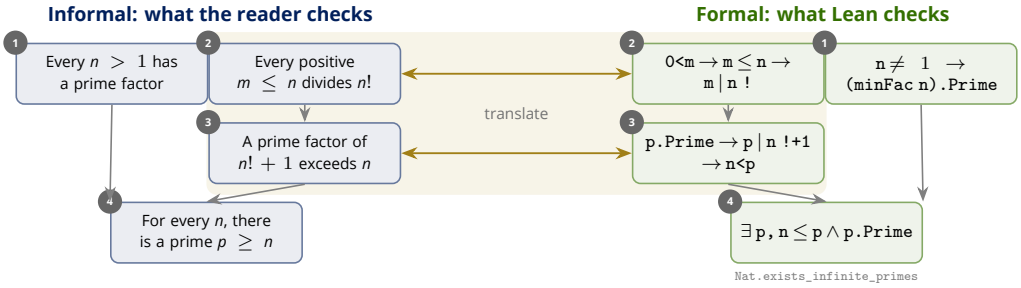
A small blueprint: there are infinitely many primes



At scale: the PFR project (Tao et al., 2023)

Image: T. Tao, *Formalizing the proof of PFR in Lean4 using Blueprint: a short tour*, blog, 18 Nov 2023 • Massot, leanblueprint • Zhu et al., LeanArchitect, ITP 2026 • Lean FRO, Verso Blueprint • Mathlib, `Nat.exists_infinite_primes`

The statement as the unit of consensus

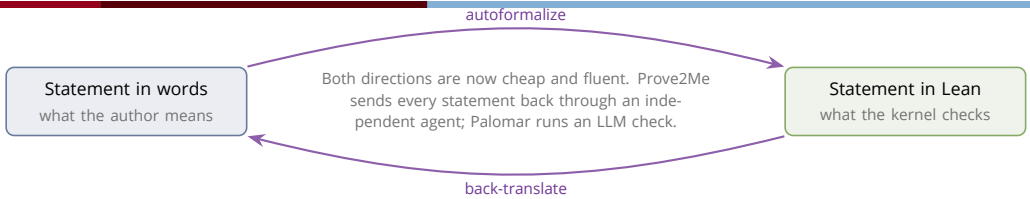


- Each node has two faces: a statement the reader understands, and one Lean checks.
- The arrows match: follow 2 to 3 and translate, or translate and then follow, and you reach the same statement.
- **Alignment:** every statement is understood directly by its audience, or supported logically by statements that are.

This is the check Navier–Stokes was missing. A Lean lemma assumed $m+5$ derivatives where the paper's needed $m+4$. Comparing the two graphs node by node points to exactly that node.

Mathlib: `Nat.minFac_prime`, `Nat.dvd_factorial`, `Nat.exists_infinite_primes`; node 3 is a lemma split out for illustration • Bastounis, Circelli & Hansen, arXiv 2610.08144

Translating statements, both ways



How reliable is it?

- **Mostly right.** An audit of OpenAI's release found about 210 of 245 formalized families consistent at the level of statements.
- **Hard to check automatically.** An LLM judge caught 63% of planted mistranslations (preprint). In general, faithfulness is undecidable.
- **Errors have shipped:** the OEIS "disproof"; Erdős #124.

Where a fluent translation can mislead

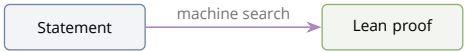
`sInf` $(\emptyset : \texttt{Set}\ \mathbb{N}) = 0$	an undefined minimum becomes 0
$(3 : \mathbb{N}) - 5 = 0$	subtraction of naturals stops at 0
$(1 : \mathbb{R})/0 = 0$	division by zero is 0
`riemannZeta 1`	has a value at the pole, by convention

Learn enough Lean to read your own statements.

Prove2Me, arXiv 2608.28433 • Tao, *Palomar*, 18 Aug 2026 • borism/openai-math-slop, issue #2 • Mohammad & Sheikh, *The Faithfulness Gap*, arXiv 2606.16541 (preprint) • Bastounis, Circelli & Hansen, arXiv 2610.08144 • formal-conjectures #5226 • Mathlib

Where formal proofs come from

From the statement alone



- AlphaProof Nexus: 9 of 353 open Erdős problems and 44 of 492 OEIS conjectures, at a few hundred dollars each.
- Works best on short, self-contained problems, where no written proof exists to follow.
- Navier–Stokes, found from the statement: about 130B tokens.

From a written proof



- Fermat's Last Theorem in 11 days, on Buzzard's project and the literature: about 6B tokens.
- The finite simple groups in 24 days, from 715 papers and books.
- A 500-page textbook: all 340 targets in about a week. Sphere packing: built on a human blueprint.

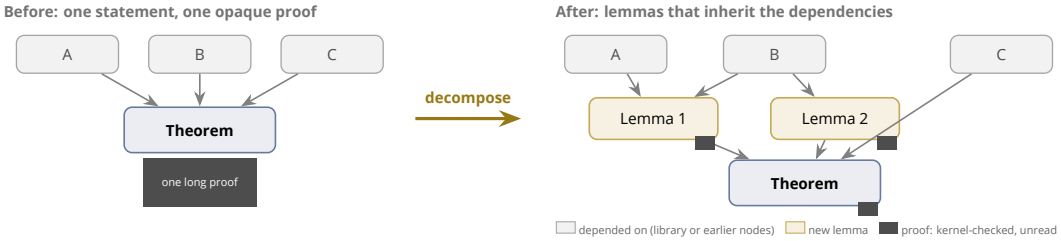
Like for like. Sketches from informal proofs raised a prover's success on competition problems from 20.9% to 39.3% (Draft, Sketch, Prove). Token counts are estimates for different problems.

And the written proof supplies the lemmas: its structure is what we decompose next.

A written proof is still the cheapest route to a formal one.

Google DeepMind, AlphaProof Nexus, arXiv 2605.22763 • Anthropic, FLT (Sep 2026) • Axiom, CFSG (Oct 2026) • Meta, arXiv 2604.03071 • AcadMathSci, Oct 2026 (B.1) • Jiang et al., arXiv 2210.12283

Decomposing the proof into interpretable lemmas



When to split a node

Proof fidelity: the lemma graph matches, not every line.

- Its statement is not a building block the audience can hold.
- Its proof is long next to its statement: something is hidden.
- Only statements are read; the route inside does not matter.

Who chooses the lemmas

- Back and forth: the human proposes nodes; the machine fills them, fails, or proposes lemmas; the human keeps those that read clearly.
- Tools extract the graph (LeanArchitect); none yet chooses interpretable lemmas.

Lemmas comprise the signature of human perspective and proof authorship.

Zhu et al., LeanArchitect, ITP 2026 • research note 02 (C, D)

Lemmas as the unit of understanding

A proof is understood when a reader can walk its graph of lemmas: read each statement, and see why it follows from those beneath it, without opening a proof.

We already work this way

- We cite results we have never read: the classification of finite simple groups rests on 715 papers and books.
- Reference works are graphs of lemmas: the Stacks Project is cited by tag.
- Many results in applied math and theoretical machine learning are not novel at the level of the lemma, but at the level of their *assembly*.

Follows established thought

- **Thurston** (1994): progress in mathematics is measured by human understanding, not by formal proof alone.
- **Lakatos** (1976): a proof decomposes a conjecture into subconjectures, or lemmas.
- **Parnas** (1972), **Wirth** (1971): use a component through its interface; refine one step at a time.

Principled blueprints, statements aligned lemma by lemma, and generated formal proofs: enough for both understanding and integrity.

Axiom, CFSG (Oct 2026) and Stacks (Nov 2026) • W. P. Thurston, *On proof and progress in mathematics*, Bull. AMS 30(2), 1994 • I. Lakatos, *Proofs and Refutations*, Cambridge UP, 1976 • D. L. Parnas, CACM 15(12), 1972 • N. Wirth, CACM 14(4), 1971

Closing

Summary

SOCIETY

A new norm, with open problems.
AI use is now routine in mathematics. Credit, careers, access and training remain unsolved.

Formal verification has advocates in all three groups.
Formalization settles correctness: it is cheap, it catches errors, and AI gains follow it.

Three criteria for trust.
Validity, statement fidelity and proof fidelity. Each has failed before.

THE LAB

Blueprints, not chat.
Agree with the machine node by node, and read your statements in Lean.

Lemmas are the unit of understanding.
Let the machine fill in the proofs; choose the lemmas that tell the story yourself.

Thoughtfully deployed, AI-assisted formalization can provide the glue between human and machine understanding.

Open problems and future directions

Open for the field

- **Credit and careers.** Who had the idea is not a formal property. Priority and hiring are unsettled.
- **Access and cost.** Only 11% of researchers have institutional funding for their main AI tool.
- **Coverage.** Libraries are thin in PDE and applied probability; requiring Lean today would leave those fields behind.
- **Training.** How do students learn to prove when the machine proves?

Future directions: things we could do

- **Formalizing our work.** We can lead the charge in including lean certificates alongside our theoretical work, in particular when we have utilized AI tools.
- **Choosing interpretable lemmas.** No tool yet picks them out of a machine proof: a learning and interpretability problem. Proof IP???
- **Checking faithfulness.** An LLM judge catches 63% of planted mistranslations (preprint).
- **Verified code.** The same question for software: does the code meet its stated specification?

Springer Nature survey, via AcadMathSci, Oct 2026 • Mohammad & Sheikh, arXiv 2606.16541 (preprint) • research notes 02-03

Math 2.0: Three principles for survival

1. **Keep control of the statement.**
   Write the blueprint yourself, and learn enough Lean to read what you are certifying.

2. **Keep your perspective.**
   Let the machine prove, but choose the lemmas, write the explanation, and pick your own problems.

3. **Keep your head up.**
   Stay aware of changes in capability and the tool landscape, and always consider best scientific practices.

Using these tools well and responsibly, Math 2.0 might not be so bad after all.

Bastani et al., PNAS • Hao et al., arXiv 2412.07727

Resources

QR code to the resources page
set `\resourcesurl` in main.tex

Slides and every link

Learn Lean

Mathematics in Lean
Theorem Proving in Lean 4
Natural Number Game
Lean Zulip

Blueprints

leanblueprint
LeanArchitect
Verso Blueprint
Tao's PFR blueprint tour

Proving and checking

Comparator
lean-lsp-mcp
Prove2Me
Palomar

Learning theory

SLT
StatsMLlib
CvxLean

Reading

Thurston, *On proof and progress in mathematics* (1994)
Lakatos, *Proofs and Refutations* (1976)
Reisinger et al., AcadMathSci report (2026) • *Proofs and Prompts*

Checking did not scale

Where	Evidence
Volume	arXiv math: 7,428 submissions in Sep 2026, against 4,252 a year earlier (+75%). OpenAI released 719 manuscripts in a single day (6 Oct)
Referees	Six probability journals paused new submissions (19 Oct – 8 Nov). Editors report they cannot find a single referee; nobody counts the declines
Formal libraries	Mathlib: 65 reviewers for 950 contributors (reported)
AI reviewing	ICLR 2026: 21% of reviews fully AI-generated
Bypassed review	Results announced on X and GitHub before any review: Navier–Stokes (8 Sep), the 719 manuscripts (6 Oct)

Production scales with compute. Expert checking scales with experts.

AcadMath5ci, Oct 2026 (§1.2, §4.7) • Nee, Zhong & Ni, arXiv 2609.04872 • Weinreich, arXiv 2608.02859 • Pangram, ICLR 2026 (self-reported) • OpenAI, 8 Sep and 6 Oct 2026

Who wrote it?

17.5%

of math preprints read as AI-written without saying so (Oct 2026).

Likely an undercount: the detector flags only half of the papers that *do* disclose. (Martin)

22.5%

of ICML 2026 reviewers used an LLM under a no-LLM policy.

Randomized trial of review policies. (Kim et al.)

No effect

AI-assisted writing grew at the same rate in journals with and without an AI policy.

5,114 journals. (arXiv 2512.06705, via AcadMathSci)

A check that depends on knowing who wrote the proof will fail.

The check has to read the proof, not the author.

K. Martin, AI use in arXiv mathematics papers (3 Nov 2026) • Kim et al., arXiv 2609.19420 • arXiv 2512.06705, cited in AcadMathSci, Oct 2026 (§1.1)

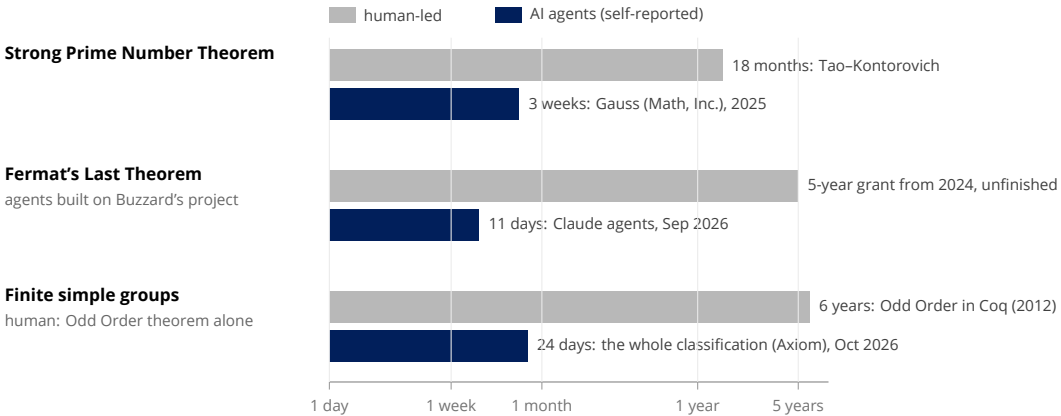
What institutions tried

Strategy	Examples	Does it hold?	Status
Restrict	arXiv: two new submissions per submitter per month (1 Oct). Six probability journals paused. *Phil. & Public Affairs* bans AI-written papers	Caps are a stopgap; the pauses ended 8 Nov. Bans depend on detection: "welcome but unenforceable"	• partial
Disclose	Leiden Declaration (endorsed by the IMU); mandatory AI statements (Springer); SIAM AI policy v2.0	Widely ignored (ICML: 22.5%); no measurable effect on AI writing	• partial
Expand	Paid, AI-assisted refereeing (Maltsev); credit for refereeing (AcadMathSci)	Needs funding, and no venue yet trusts AI review with the deciding vote	• open
Verify	Palomar, a registry of Lean-verified results (18 Aug). A proposed journal requiring Lean proofs (Guo, 17 Oct). AcadMathSci: assess correctness "utilising AI and formalisation"	Scales with production and does not depend on who wrote the proof. Still prototypes	• open

Only verification scales with production, and it is the least built.

AcadMathSci, Oct 2026 (§1.2, §4.6, C.1) • Times Higher Education, May 2026 • Kim et al., arXiv 2609.19420 • Maltsev, Proofs & Prompts • Tao, blog, 18 Aug and 17 Oct 2026

Formalization got cheap



Log scale. The AI-era figures come from the companies; the finite-simple-groups build has not been independently reproduced.

Math, Inc., Strong PNT (2025) • Anthropic, FLT (Sep 2026) • Axiom, CFSG (Oct 2026) • Gonthier et al., ITP 2013 • research note 01, §1

Certificates catch what review missed

OpenAI's 719 manuscripts

Sep–Oct 2026 • independent audit

- 3 withdrawn within days after a sign error. **All three were outside the Lean coverage.**
- An audit of the 245 formalized families found no theorem shown false.
- But 21 headline results labelled "(Lean)" had nothing formalized: check the certificate, not the label.

Finite simple groups

Axiom, Oct 2026 • self-reported, build not yet reproduced

- 34 deviations from a published record that experts had checked for 40 years.
- A false equality in a published quasithin lemma: a counterexample in $\mathrm{SL}_2(5)$.
- An unjustified step in the Unbalanced Group Theorem (1986), repaired by a new route.

The Stacks Project

Axiom, Nov 2026 • self-reported

- Dozens of statements that fail in the stated generality: missing hypotheses, edge cases in positive characteristic.
- Each Lean declaration links to its Stacks tag, so a human can check one statement at a time.

borism/openai-math-slop, issue #2 • OpenAI release record (Wikipedia) • Axiom, CFSG (Oct 2026) and Stacks (Nov 2026) • research note 01, §§2–3, 5

The paper and the code disagree

Navier–Stokes (Bastounis, Circelli & Hansen, Oct 2026)

- The paper's estimate (8.19) needs $m+4$ derivatives; the matching Lean theorem assumes $m+5$, and the pattern repeats.
- The paper's pressure bound depends on one quantity; the Lean bound depends on two, and is proved by a different route.
- They do not claim the paper is wrong.

OpenAI's 719 manuscripts (community audit of 245 formalized families)

- About 210 consistent at the level of statements; no theorem shown false.
- 21 headline results marked "(Lean)" were not formalized; 5 formalized only a supporting lemma.

> *Lean acceptance (compilation) of translation does not imply semantic preservation.*
>
> — Bastounis, Circelli & Hansen, arXiv 2610.08144

The drift was found because there was formal code to hold the paper against.

A certificate proves *a* theorem. Whether it is *the* theorem is a human question.

Bastounis, Circelli & Hansen, arXiv 2610.08144 • borism/openai-math-slop, issue #2 • research note 01, §5

Credit, disclosure and confidentiality

Problem	Case	Response	Status
Credit	AI reuses known ideas without attribution: 8 of Aletheia's 13 meaningful Erdős results were already in the literature. Kolda: "Human–AI scrapple"	Leiden Declaration (Jun 2026, endorsed by the IMU): human responsibility, attribution	• partial
Priority	Navier–Stokes priority dispute (Córdoba & Martínez-Zoroa; Buckmaster & Alpöge)	AGMAI release guidelines (29 Sep): review before announcement, release prompts. Labs withheld prompts	• open
Disclosure	1 in 6 math preprints read as AI-written without saying so. At ICML 2026, 22.5% of reviewers used an LLM under a ban	Disclosure statements (arXiv, Springer, Leiden); SIAM AI policy v2.0	• partial
Confidentiality	Referees uploading manuscripts to AI tools	LMS: misconduct. SIAM v2.0: "non-delegation of scholarly judgement"	• partial

The rules already bind individuals; none of them can be checked from the paper alone.

Reisinger et al., AcadMathSci, Oct 2026 (A.3, C.1, C.4, C.5) • Kolda, Silicon Reckoner, Feb 2026 • K. Martin, *AI use in arXiv mathematics papers* (3 Nov 2026) • Kim et al., arXiv 2609.19420

Training, access and narrowing

Problem	Case	Response	Status
Training	Students given unrestricted GPT-4 scored 17% lower once access was removed; a tutor with guardrails largely avoided the harm (Bastani et al.)	Vivas and in-person exams; AI-permitted work graded on formulation and validation	• partial
Access	Only 11% of researchers have institutional funding for their main AI tool (Springer Nature, 10,480 researchers). Navier–Stokes cost an estimated £4.9M	Token costs eligible on grants (proposed); SAIR Open Math OS: free open models for academics	• partial
Narrowing	AI adoption shrinks topic range by 4.6% and engagement by 22% (Hao et al., 41M papers). Agendas drift toward “problems easily amenable to current AI”	Reward understanding and long-term contribution over volume (AcadMathSci)	• open

Reisinger et al., AcadMathSci, Oct 2026 (§1.6, §3.2, §3.3, §4.3, B.1) • Bastani et al., PNAS • Hao et al., arXiv 2412.07727

Corpus and scoring method

[– min] Details behind the opinion map

Backup slides

[– min] Material for questions

References

[– min] Sources